MSG No. 192.251 $P6/m'c'c'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `622`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `6/m'..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,4,5,6,13,14,15,16,17,18]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[7,8,9,10,11,12,19,20,21,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{4}\right]$	`[1,3,5,10,11,12]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{4}\right]$	`[2,4,6,7,8,9]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{3}{4}\right]$	`[13,15,17,22,23,24]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{3}{4}\right]$	`[14,16,18,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `-6'..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, 0\right]$	`[1,3,5,14,16,18]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, 0\right]$	`[2,4,6,13,15,17]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}\right]$	`[7,8,9,22,23,24]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}\right]$	`[10,11,12,19,20,21]`

Table 5: Wyckoff site: `4e`, site symmetry: `6..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3,4,5,6]`
2	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[7,8,9,10,11,12]`
3	$[0,\, 0,\, -z]$	`[13,14,15,16,17,18]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[19,20,21,22,23,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `222`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, \frac{1}{4}\right]$	`[1,4,7,11]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{4}\right]$	`[2,5,9,10]`
3	$\left[0,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,6,8,12]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[13,16,19,23]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[14,17,21,22]
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[15,18,20,24]

Table 7: Wyckoff site: 6g, site symmetry: 2/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,13,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,14,17]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,15,18]
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[7,11,19,23]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[8,12,20,24]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,21,22]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2} - z\right]$	[7,8,9]
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[10,11,12]
5	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[13,15,17]
6	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[14,16,18]
7	$\left[\frac{1}{3}, \frac{2}{3}, z + \frac{1}{2}\right]$	[19,20,21]
8	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6]
4	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[7,11]
5	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[8,12]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[9,10]
7	$\left[\frac{1}{2}, 0, -z\right]$	[13,16]
8	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[14,17]
9	$\left[0, \frac{1}{2}, -z\right]$	[15,18]
10	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[19,23]
11	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[20,24]
12	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[21,22]

Table 10: Wyckoff site: `12j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[x,\ x,\ \frac{1}{4}\right]$	`[2,10]`
3	$\left[0,\ x,\ \frac{1}{4}\right]$	`[3,8]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[5,9]`
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[6,12]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[13,19]`
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[14,22]`
9	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[15,20]`
10	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[16,23]`
11	$\left[x,\ x,\ \frac{3}{4}\right]$	`[17,21]`
12	$\left[0,\ x,\ \frac{3}{4}\right]$	`[18,24]`

Table 11: Wyckoff site: `12k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,9]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[-x,\ -2x,\ \frac{1}{4}\right]$	`[4,7]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,10]`
6	$\left[2x,\ x,\ \frac{1}{4}\right]$	`[6,8]`
7	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[13,23]`
8	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[14,21]`
9	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[15,24]`
10	$\left[x,\ 2x,\ \frac{3}{4}\right]$	`[16,19]`
11	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[17,22]`
12	$\left[-2x,\ -x,\ \frac{3}{4}\right]$	`[18,20]`

Table 12: Wyckoff site: `12l`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,16]`
2	$[x-y,\ x,\ 0]$	`[2,17]`
3	$[-y,\ x-y,\ 0]$	`[3,18]`
4	$[-x,\ -y,\ 0]$	`[4,13]`
5	$[-x+y,\ -x,\ 0]$	`[5,14]`
6	$[y,\ -x+y,\ 0]$	`[6,15]`
7	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[7,23]`
8	$\left[y,\ x,\ \frac{1}{2}\right]$	`[8,24]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[9,22]`
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[10,21]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[11,19]`
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[12,20]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[x-y,\ x,\ z]$	`[2]`
3	$[-y,\ x-y,\ z]$	`[3]`
4	$[-x,\ -y,\ z]$	`[4]`
5	$[-x+y,\ -x,\ z]$	`[5]`
6	$[y,\ -x+y,\ z]$	`[6]`
7	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	`[7]`
8	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[8]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	`[9]`
10	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	`[10]`
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[12]`
13	$[-x,\ -y,\ -z]$	`[13]`
14	$[-x+y,\ -x,\ -z]$	`[14]`
15	$[y,\ -x+y,\ -z]$	`[15]`
16	$[x,\ y,\ -z]$	`[16]`
17	$[x-y,\ x,\ -z]$	`[17]`
18	$[-y,\ x-y,\ -z]$	`[18]`
19	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	`[19]`
20	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[20]`
21	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	`[21]`
22	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	`[22]`
23	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	`[23]`
24	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[24]`